

UNSUPERVISED MACHINE LEARNING

MODULE 3:

SELECT A CLUSTERING ALGORITHM

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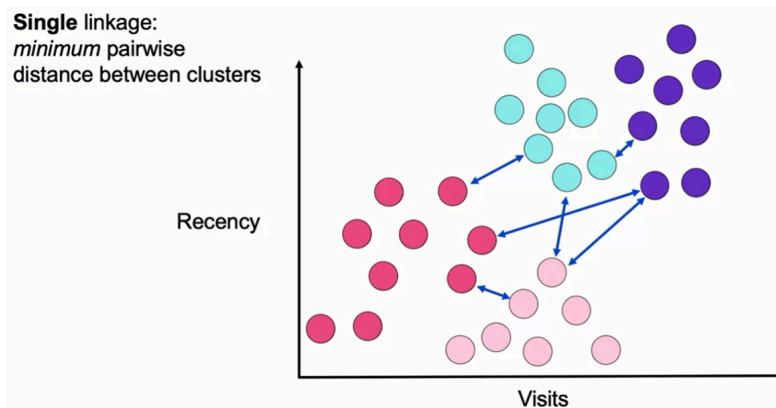
Hierarchical Agglomerative Clustering (HAC)

Concept

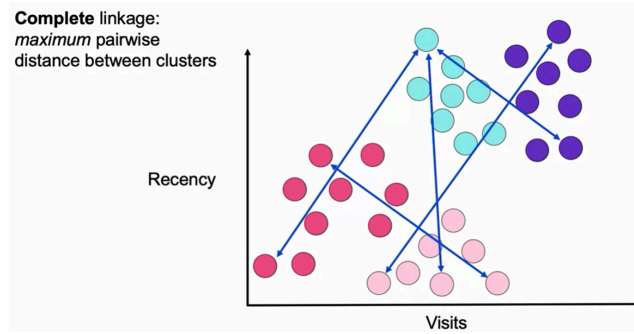
- HAC successively **merges the closest points or clusters** until only one cluster remains.
- A **stopping criterion** is needed to prevent merging everything into one cluster.
- The algorithm depends on **distance metrics** and **linkage methods** to determine how clusters are joined.

Linkage Methods

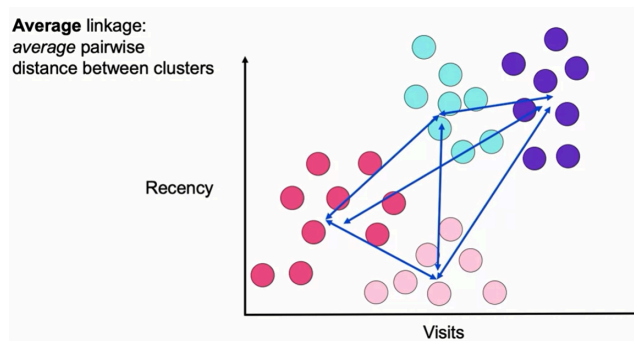
- **Single linkage:** Minimum distance between points of clusters.
 - Pro: Maintains clear separation.
 - Con: Sensitive to noise and outliers.



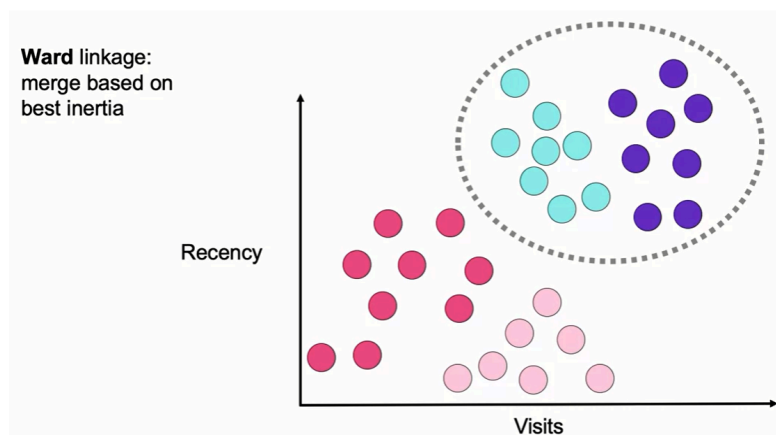
- **Complete linkage:** Maximum distance between points of clusters.
 - Pro: Handles overlapping clusters better.
 - Con: May break large clusters apart.



- **Average linkage:** Average distance between all pairs of points.
 - Balanced approach between single and complete.



- **Ward linkage:** Merges clusters that **minimize the increase in inertia** (sum of squared distances).
 - Pro: Similar to k-means, balances variance well.
 - Con: Sensitive to cluster size.



Stopping Rule

- Stop merging when the **average inter-cluster distance** exceeds a chosen threshold.

```
from sklearn.cluster import AgglomerativeClustering

agg = AgglomerativeClustering(
    n_clusters=3,
    affinity='euclidean',
    linkage='ward')
agg = agg.fit(X1)
y_pred = agg.labels_
```

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

Concept

- Groups points based on **density** rather than distance to a centroid.
- Does not require specifying the number of clusters.
- Can identify **outliers (noise)** that do not belong to any cluster.

Core Parameters

- **Epsilon (ϵ):** Radius of the neighborhood around a point.
- **Min_samples (n_clu):** Minimum number of points required to form a dense region (a “core point”).

Point Categories

- **Core Point:** Has $\geq \text{min_samples}$ points within ϵ .
- **Border Point:** Lies within ϵ of a core point but has fewer neighbors.
- **Noise Point:** Not within ϵ of any core point.

Visualization Logic

- Randomly choose a point $\rightarrow$ check neighbors within $\epsilon \rightarrow$ expand cluster if density is met $\rightarrow$ repeat for unvisited points.

```
from sklearn.cluster import DBSCAN

db = DBSCAN(eps=3, min_samples=2)
db.fit(X)
labels = db.labels_ # -1 = noise
```

Strengths

- Detects **arbitrary-shaped** clusters.
- Handles **noise/outliers** naturally.
- No need to predefine the number of clusters.
-

Weaknesses

- Sensitive to ϵ and `min_samples` choice.
- Poor performance on **clusters with varying densities**.
- Difficult parameter tuning in **high dimensions**.

Mean Shift Clustering

Concept

- Similar to k-means, but instead of centroids, it finds **density peaks (modes)**.
- Points “shift” toward regions of **highest density** within a **window (bandwidth)**.

Algorithm Steps

1. Select a point and a **window** around it.
2. Compute **weighted mean** of points in that window.
3. Shift the window to the new mean.
4. Repeat until convergence (no further shift).
5. Points converging to the same mode form a cluster.

Formula for Weighted Mean

$$m(x) = \frac{\sum_{x_i \in N(x)} K(x_i - x) x_i}{\sum_{x_i \in N(x)} K(x_i - x)}$$

- K is typically an **RBF (Gaussian)** kernel, weighting closer points more

```
from sklearn.cluster import MeanShift

ms = MeanShift(bandwidth=2)
ms.fit(X1)
y_pred = ms.predict(X2)
```

Strengths

- No need to define the number of clusters.
- Robust to outliers.
- Works well for density-based structures.

Weaknesses

- Depends heavily on bandwidth choice.
- Computationally expensive ($O(m \cdot n^2)$).
- Struggles with large datasets.

Key Takeaways

Algorithm	Predefined Clusters	Handles Noise	Shape Flexibility	Key Parameter	Scalability	Common Use
K-Means	Yes	No	Spherical only	k	Excellent	Simple, fast grouping
Mean Shift	No	Yes	Moderate	Bandwidth	Slow	Unknown cluster count

Hierarchical (Ward)	Optional	Somewhat	Moderate	Linkage type	Medium	Subgroup discovery
DBSCAN	No	Yes	High	ϵ , MinPts	Medium	Outlier detection, non-spherical shapes

General Guidance

- **K-Means:** When cluster count is known and data is balanced.
- **Mean Shift:** When clusters are unknown; smaller datasets.
- **Hierarchical:** When hierarchical relationships matter.
- **DBSCAN:** When clusters are irregular and contain noise.